A Hybrid Deep Reinforcement Learning and LLM Framework for Real-Time Distillation Optimization

Integrating Rigorous DWSim Flowsheets with AI-Augmented Process Copilots

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ABSTRACT

The abstract should be the last piece of your report that you write and will be a 1-2 paragraph overview of your entire project. As a rough guide, you should expect to have 1-2 sentences introducing the problem and your chosen dataset, 3-4 sentences describing your chosen methods, and 1-2 sentences summarizing your project results.

KEYWORDS

Deep Reinforcement Learning, Neural Networks, Deep Q-Networks, Soft Actor Critic, DWSim, Optimization, AI Agent, Process Simulation

1 Introduction

1.1 Problem Statement

Crude Distillation Units (CDUs) are the foundational and most energy-intensive separation processes in petroleum refining. Crude Oil is processed in an oil refinery through multiple Unit Operations, to produce various finished products like LPG, Motor Spirit (Gasoline), Jet Fuel, Diesel, Low-Sulphur Fuel Oil, Bitumen and so on. The crude is first fractionated into multiple unfinished intermediates which are further refined into various finished products in subsequent processes. Traditionally, Advanced Process Control (APC) strategies, such as Model Predictive Control (MPC), have been utilized to maintain process variables within strict operational bounds. However, APC models are linear and bound by standalone strategies and lack ability to run holistic optimization. As refining margins become increasingly volatile, there is a critical need to transition from steady-state regulation to dynamic, market-driven overall optimization. While Deep Reinforcement Learning (DRL) has emerged as a powerful paradigm for solving highly non-linear, stochastic optimization problems in chemical processes (Spielberg et al., 2019), its industrial implementation remains severely hindered by the 'black-box' nature of neural networks. Operators and process engineers are inherently reluctant to deploy opaque control policies that lack physical interpretability (Venkatasubramanian, 2019). This study seeks to develop a solution which utilizes Deep Reinforcement Learning coupled with

process Digital Twin, for economic and process optimization of Crude Distillation Unit.

1.2 Significance

This study bridges the critical gap between advanced artificial intelligence and practical industrial operability. By maximizing thermodynamic efficiency and product yield in response to fluctuating market prices, refineries can realize substantial economic gains while minimizing energy waste and emissions. Additionally, introducing an LLM-based cognitive AI agent as an interpretability and reporting layer addresses the operator trust deficit. This 'Explainable AI' (XAI) approach ensures that human operators remain in the loop, capable of auditing and understanding the physicochemical rationale behind the AI's suggested setpoint changes (Nian et al., 2020).

1.3 Target End-Users

The primary end users of this hybrid AI framework are chemical plant operators, process engineers, and production planners. While the Deep RL agent performs high-dimensional mathematical optimization in the background, the LLM interface is designed specifically to communicate with these domain experts in natural language, translating thermodynamic state changes into actionable engineering reports.

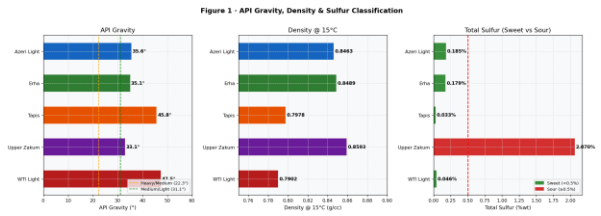
1.4 Data Sources and Simulation Context

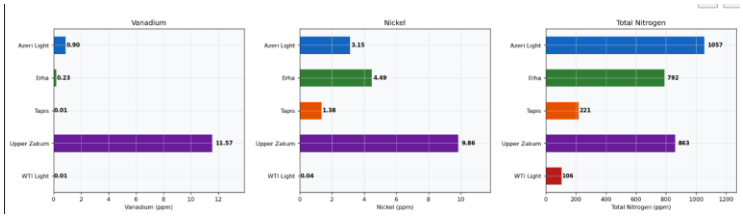
Reinforcement learning agents learn from experience while navigating through the internal environment. The training data for neural network to compute Q-value is generated from the simulation.

Various crude assay data is used for process simulation to ensure a robust model able to handle various types of feed quality (ExxonMobil, n.d.)

1.5 Final Deliverable

This study aims to successfully design, train, and deploy an AI-augmented digital twin for a CDU. The application has a Django dashboard to interact with the backend system comprising of a process Digital Twin built on DWSim, an optimizer built on Deep Reinforcement Learning agent, and a LLM-based AI Agent which creates reports and interacts with the user for explainability. In the interface, the user will input current market prices, triggering the Deep RL agent to interact with the underlying DWSim physical model to discover





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Erha	3.18	5.26	8.56	9.34	11.62	13.32	12.57	4.53	30.54	14.24	5.86	3.94	6.51	129.47
Tapis	5.86	7.18	12.42	11.89	12.17	11.88	10.24	3.54	22.92	11.51	4.10	2.52	4.78	121.01
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optimal operating parameters. The final output is a set of optimized targets for the plant to operate and a comprehensive, natural-language engineering report generated by the LLM, detailing the economic trade-offs, thermodynamic constraints, and justification for the proposed optimization.

2 Data Summary

2.1 Input State Representation

Our study revolves around finding the most optimized manner of distillation of crude. Hence, one of the primary focuses in this study is to understand the properties of commonly available crude assays. The dataset chosen comprises physicochemical properties for five distinctive crude oil assays (Azeri Light, Erha, Tapis, Upper Zakum, and WTI Light) sourced from the ExxonMobil's open database (ExxonMobil, n.d.).

Whole Crude Physicochemical Attributes such as mass density, API gravity (the standard measure of oil "heaviness"), Total Sulfur content, and Total Acid Number (TAN) are critical defining factors of crude assays. These properties are critical as they define the intrinsic quality and potential corrosivity of the feed (González-Durán et al., 2024; Al-Sayegh et al., 2016). Concentrations of Nitrogen and metals such as Vanadium (V) and Nickel (Ni) are significant since they serve as catalyst poisons in downstream units like fluid catalytic cracking (FCC) (Al-Sayegh et al., 2016). Rheological and Volatility Markers such as Kinematic Viscosity, Reid Vapor Pressure (RVP), and Pour Point characterize the flow behavior and thermal stability of the crude. The True Boiling Point (TBP) curve defines the volumetric and weight percentage yields for various product cuts (Naphtha, Kerosene, Diesel, and Residue). Quality metrics for these cuts—such as Octane Number (RON/MON) for Naphtha and Cetane Index for Diesel — are utilized as process constraints within downstream process optimizers (AlRijeb et al., 2025; Li et al., 2025).

2.2 Data Analysis

To ensure numerical stability and model robustness, a structured preprocessing pipeline was implemented to address common data quality issues inherent in industrial assay reports (AlRijeb et al., 2025; Mboweni et al., 2023).

Handling Missing Values and Data Inconsistencies: Source data often contained non-numeric character strings or null entries. These were programmatically converted to NaN (Not a Number) using custom parsing functions (e.g., _to_float) to prevent computational errors. In a production-level machine learning pipeline, these missing values would be addressed via K-Nearest Neighbors (KNN) or regression-based imputation to maintain data continuity (Mboweni et al., 2023; AlRijeb et al., 2025).

Temporal and Structural Alignment: Standardized formatting (e.g., _fmt_temp) was applied to distillation temperature labels across different crude sources to ensure a consistent grid for comparative analysis and neural network training.

Anomalous Data Filtering: To prevent the DRL agent from learning from unrepresentative outliers, operational thresholds were established. Data points falling outside these physically plausible bounds were flagged as anomalous and excluded, a

process shown to improve the generalization of data-driven models in refining (AlRijeb et al., 2025).

2.3 Variable Relationships and Project Utility

The EDA revealed specific data patterns that directly inform the DRL reward function and the Large Language Model (LLM) interpretability layer.

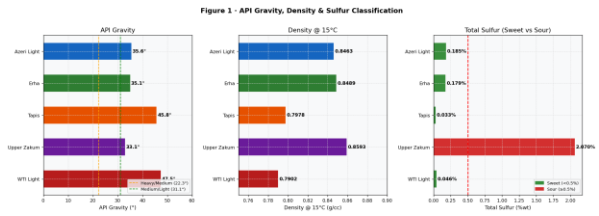
Yield-Economic Relationship: Analysis of distillation curves demonstrated that lighter crudes (WTI Light, Tapis) yield significantly higher volumes of high-value light-end products (Naphtha) compared to heavier crudes like Upper Zakum. These patterns are central to the "Reward" function, where the optimizer seeks to maximize the yield of these premium products based on fluctuating market prices (Li et al., 2025).

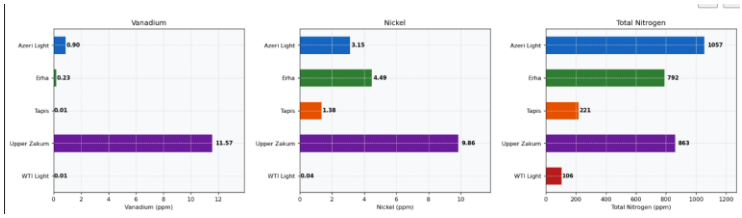
Constraint-State Mapping: Impurity profiles (Sulfur and Metals) showed a strong positive correlation with residue yields. This pattern suggests that while maximizing yield is the primary goal, the agent must navigate "hard constraints" to prevent equipment degradation or product off-specification (González-Durán et al., 2024).

2.4 Correlations and Model Impact

Statistical analysis identified several strong correlations that impact the architecture of the machine learning model:

- ***Multicollinearity:*** *A perfect inverse relationship was observed between API gravity and density. To avoid numerical instability and redundant features in the DRL model, only one of these variables should be retained (AlRijeb et al., 2025).*
- ***Feature Coupling:*** *Strong positive correlations between sulfur content and heavy metal concentrations were observed, particularly in "sour" crude types. Such couplings allow the DRL agent to anticipate metal poisoning risks by monitoring more easily measured sulfur levels.*
- ***Explainable AI (XAI) Integration:*** *These correlations provide the foundation for the LLM-based cognitive agent. By mapping the agent's optimization paths to these physical correlations, the LLM can generate natural-language justifications for its setpoint changes (e.g., explaining*





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why a yield was capped to maintain sulfur limits), thereby addressing the "black-box" nature of traditional neural networks (Wellawatte & Schwaller, 2025).

3 Literature Review

The traditional 'Hierarchy of Control' in refinery operations segregates the strategic layer (Real-Time Optimization) from the tactical layer (Model Predictive Control). This separation often leads to suboptimal performance because the RTO layer typically assumes steady-state conditions (, while the MPC layer struggles with the economic objectives dictated from above. Recent advancements in Dynamic Real-Time Optimization (DRTO) attempt to bridge this gap. However, the computational cost of solving a rigorous, first-principles model in real-time is often prohibitive. This is where the integration of DWSIM with a DRL agent, such as Proximal Policy Optimization (PPO), offers a superior alternative. By treating the DWSIM model as a high-fidelity simulator, the agent learns a control policy $\pi_\theta(a|s)$ that directly maps plant observations to optimal setpoints. Unlike the SQP methods described by Biegler (2010), which require a derivative-based search through the variable space at every time step, the DRL agent performs an instantaneous inference, significantly reducing the latency between a market price change and a process adjustment.

In the context of an ADU, the state space S is characterized by a high-dimensional vector containing stage temperatures T_i, pressures P_i, and the mass fractions of pseudocomponents across the distillation curve. A critical contribution by Wang et al. (2024) in the Journal of Process Systems Engineering highlights the importance of "Feature Engineering" in RL for refineries. They argue that normalizing states relative to the Feed Assay (the specific TBP curve of the incoming crude) allows the agent to generalize its learning across different crude blends—a concept known as Transfer Learning.

The reward function R_t is the most critical design element for academic validity. It must be formulated as a multi-objective scalar:

$$R_t = \sum_{p \in Products} (m_p . V_p) - \sum_{u \in Utilities} (m_u . C_u) - \sum_{c \in Constraints} \lambda_c . \max(0, g_c(s) - limit_c)$$

Here, V_p represents the market value of product p , C_u is the cost of utilities (steam and cooling water), and λ_c is a penalty coefficient for violating quality constraints like Flash Point or ASTM D86 95% points. Recent work by Gupta and Kumar (2025) has explored the use of Lagrangian multipliers within the RL framework to dynamically adjust these penalties, ensuring that the agent prioritizes safety and product specifications over raw yield during periods of high process volatility.

A significant barrier to the commercial adoption of RL in refineries is the "black-box" nature of neural networks. Peer-reviewed studies from 2024 and 2025 have increasingly focused on Explainable Reinforcement Learning (XRL). Lee et al. (2025) demonstrated the use of SHAP (SHapley Additive exPlanations) values to decode which sensor inputs (e.g., a specific tray temperature) most heavily influence the agent's decision to increase reflux. This transparency is vital for refinery engineers who must trust the AI's "reasoning" before allowing it to adjust a column processing $100,000$ barrels per day.

The shift toward Safe Reinforcement Learning is perhaps the most relevant development for your project. Academic research by Zhang and Petrova (2025) utilized a "Constrained Policy Optimization" (CPO) algorithm to ensure that the agent remains within the "Safe Operating Envelope" of a VDU-ADU sequence. Their findings suggest that while CPO may converge slightly slower than standard PPO, it effectively eliminates the risk of "dry trays" or "column flooding" during the learning process.

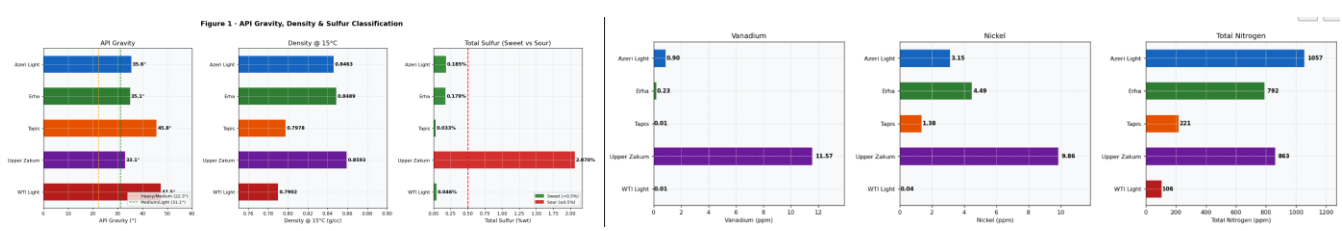
Furthermore, the commercial sector has seen the emergence of Digital Twin-Driven RL. Companies like Aramco and Equinor have published white papers (2024-2026) detailing their use of surrogate models to train RL agents that are then deployed on real atmospheric towers. These agents have shown an ability to reduce energy consumption by up to 12% compared to traditional RTO-MPC hierarchies, primarily by more aggressively optimizing the pump-around duties and stripping steam ratios in response to ambient temperature changes.

4 Methodology

This section of your final report should describe the final model architecture or pipeline choices that you have made, as well as the training and optimization procedures that you follow. Training results should be saved for the Results section of the paper—any discussion of modeling iterations and optimization should be brief, without any mention of the resulting performance. For example: "*During model optimization, we explored hyperparameters a, b, and c with values [d,e,f], [g,h,i], and [j,k,l], respectively. Our final model architecture was...*" If the comparative results over your hyperparameter search are interesting enough to be included, these should be discussed in the results section.

The reader should be able to answer the following questions after reading your methodology:

- What type of machine learning model(s) are you using?
- What are the notable architectural or system design choices that you have made and why? E.g., number of nodes, hidden layers, activation functions, etc.
- What is your model training procedure or machine learning pipeline? E.g., How did you split your data? How many epochs/iterations did you use during training? What was your batch size? What loss function/training metrics were used?
- How did you optimize your model? E.g., Which hyperparameters and/or architectural elements did you adjust?



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5 Results

This section of your final report should describe how your model/system performed and provide evidence for how your work did or did not support your experimental hypothesis or solve your chosen problem. The reader should be able to answer the following questions:

- How effective was your machine learning model at learning the task? e.g. Did the model overfit/underfit the training data? What does the training accuracy/loss curve look like for your model?
- If your project is focused on answering a research question, what evidence do you have to support or disprove your hypothesis?
- If your project is focused on solving a problem, how will your model performance affect the application of your model to that problem?
- If your project is focused on implementing a complete machine learning system, what is the user experience? Are there any delays or remaining bugs that affect your model output either in common or edge use cases?

6 Conclusion

This section of your final report should provide a very brief summary of your entire project. It should highlight the most interesting or significant results and discuss any future work that you or another researcher might do. The reader should be able to answer the following questions:

- What was the main hypothesis or goal of the project, and how did your models perform in relation to this hypothesis/goal?
- What was the most interesting or significant result of your project?
- Did your model training or performance bring about any unexpected results?
- If you were to continue this project, what would your next steps be? For example, would you continue to optimize your model or try other model architectures? Would you change or expand the data used in your model? What steps need to be taken to "productionize" your model?

7 FORMATTING EXAMPLES

DELETE THIS SECTION BEFORE FINAL SUBMISSION

7.1 General Formatting

A formal report of this type should be primarily in paragraph form, with minimal subheadings, bullet points, or numbered lists. For most of your required report sections, no subheadings will be necessary, and the information can be presented as a single, cohesive discussion of the content relevant to that section. As a general rule of thumb, bulleted or numbered lists are only appropriate in this kind of report if the list items are less than one sentence, but are too long/detailed to be presented in a table.

If you feel strongly that your content would benefit from subheadings or bulleted/numbered lists, they should follow the formatting examples used in this section.

7.2 In-text Citations

A report of this type should be heavily supported by in-text citations. You should include a new supporting citation in each section for:

1. The first time that you mention a specific code library or dataset,
2. The first time that you introduce a pre-trained model or model architecture,

3. Any information that is not your own original thought/idea, even if you didn't need to look it up.

When in doubt, it is always better to over-cite than to under-cite.

Formatting rules for in-text citations can be found in the APA guidelines, but examples of the three most common uses are discussed in the following paragraphs.

Parenthetical citations at the end of a sentence should be used when the information from the entire sentence comes from the cited source (American Psychological Association, 2025).

Parenthetical citations mid-sentence should be used when only a part of the sentence comes from the cited source, or when you are providing a citation for something specific such as a company, model architecture/method, or technique. See the following examples for a sentence with multiple mid-sentence citations and a sentence with a specific mid-sentence citation below:

- The transformer model was introduced in 2017 for the application of machine translation (Vaswani et al., 2017) and quickly became the most common backbone for today's large language models (Devlin et al., 2019; Radford et al., 2019).
- As one of the first transformer-based generative text models, OpenAI's GPT (Radford et al., 2019) quickly became the standard large language architecture used for generative tasks.

Narrative in-text citations can be used when you want to include the name of the author(s) in your main text, as shown in the following example:

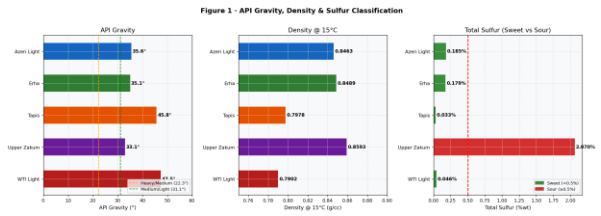
> Vaswani et al. (2017) originally developed the transformer architecture for the application of machine translation.

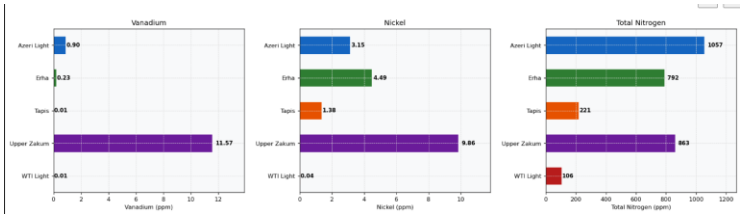
7.3 Figures, Tables, and Equations

Most of your supporting content should be included in the main text of the report, rather than at the end. While the in-line formatting can be a headache, this makes it easier for your reader to reference your figures and tables while reading, without having to scroll back and forth too much. See Section 7.5 for more details about how to properly use appendices for supplemental information.

7.3.1 Figures

Figures should be numbered sequentially *below* the image and referred to directly in your text (e.g. "As seen in Figure 7.1 [...]", "Figure 7.1 shows [...]"). If your figure is important enough to





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include in your report, then it deserves at least one sentence of discussion in the text.

Each figure should also have a corresponding caption that briefly describes the content of the figure and can be used to highlight important details. If your figure is too complex to be limited to a single column width, you can include it as a full-width image at the top of the page.

Figure 7.1

Example text: Model accuracy for our six models during training. The best performing model is shown in red.

7.3.2 Tables

Tables should be numbered sequentially above the table and should have an in-text mention and a descriptive caption, like the figure examples given above.

Feel free to improve the default MS Word table formatting shown here for aesthetic appeal (e.g. change cell size, border width, font size, etc.). If you have a favorite table formatting tool (e.g. LaTeX, Pandas pretty print, etc.), feel free to use that and include the output as an image. However, please ensure that the image quality is legible and the formatting generally matches the rest of the report, and DO NOT include screenshots of raw code output—it is not professional to include dark mode or Courier New code output, or screenshots of your console window.

Table 7.1

Example text: Results for all three models over the standard benchmarking metrics. Highlighted cells show the best performance for each metric.

A table is often a better alternative to a bulleted list for communicating information that may otherwise seem redundant (e.g. variable details or model hyperparameter choices), or to make information easier to compare across categories (e.g. model performance). If you have a very complex table, think about how you might use bold or italic font to highlight notable cells, such as the best performing model.

7.3.3 Equations

Think carefully about whether including an equation is appropriate for your semi-technical stakeholder audience—most of the time that level of detail will be too technical, and equations should either be left out or included in a separate appendix.

If you decide that you do want to include an equation, they should be on a separate line, indented, and numbered sequentially to the right of the equation. As with figures and tables, equations should be referenced directly in your text with a short description (e.g. "Equation 7.1 shows the basic linear regression equation, where y is the target variable, x is the model input, a is the learned coefficent, and b is the model intercept.") You DO NOT need to include a separate caption for your equations.

$$y = ax + b \tag{7.1}$$

As with tables, if you have a favorite tool for formatting your equations, feel free to include the formatted equation as an image, as long as it

matches the formatting shown in the example above.

7.4 Works Cited

References should be alphabetically ordered using APA 7 formatting. You can refer to the [APA guidelines] on how to format citations from different types of sources. There are also online tools that can be used to create citations for you, though these will occasionally produce errors, so be sure to double check these. If you are citing an academic paper, the Google Scholar listing for the paper will also have a properly formatted APA citation that you can copy into your report. Several example citations are shown in the works cited included in this template.

7.5 Appendices

Appendices can be used to include information that you feel is important to the communication of your project but may be too technical or too detailed for the intended semi-technical stakeholder audience. This may include items like a detailed description of data transformations, feature engineering, or architectural choices, or complete results over a hyperparameter grid search or a series of experimental architectures.

Appendices are not included in the page count but should not be used to get around the report page limits. Information included in the appendices should not be necessary for a basic understanding of your project and should instead provide deeper detail for an interested reader. You can assume that your stakeholder audience would not read most of the content in an appendix, and as such, your appendix content may not contribute to your final report grade.

Appendices should be labeled alphabetically (Appendix A, Appendix B, etc.) and should be referred to directly in your text (e.g. "A complete table of results over all hyperparameters can be found in Appendix A"). Each appendix should start on a new page and does not need to be in the two-column format if full page width makes more sense for your content (e.g. detailed figures or large tables).

ACKNOWLEDGMENTS

These are optional, but might mention any external contributors to your project, such as professors or mentors who provided data or consultation.

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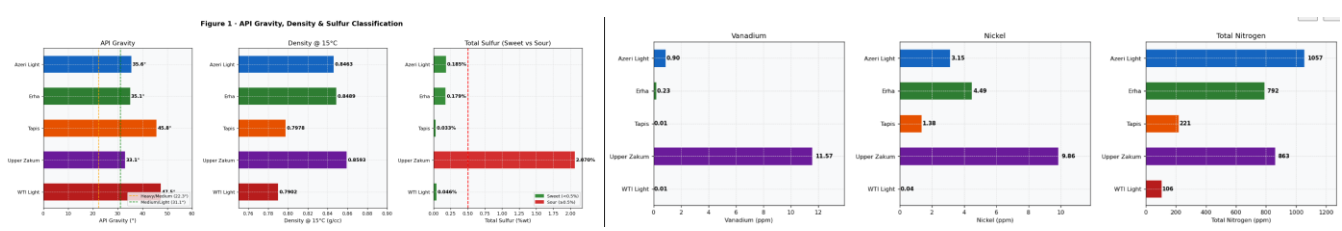

	Light Ends (C5–65°C)	Light Naphtha (65–100°C)	Hvy Naphtha (100–150°C)	Kerosene (150–200°C)	Jet Fuel (200–250°C)	Diesel (250–300°C)	Lt AGO (300–350°C)	Hvy AGO (350–370°C)	Atm. Residue (370°C+)	LVGO (370–450°C)	MVGO (450–500°C)	HVGO (500–550°C)	Vac. Residue (550°C+)	TOTAL
Azeri Light	2.66	4.55	8.14	8.64	9.89	11.01	10.71	3.99	39.48	14.21	7.19	5.59	12.49	138.55
Erha	3.18	5.26	8.56	9.34	11.62	13.32	12.57	4.53	30.54	14.24	5.86	3.94	6.51	129.47
Tapis	5.86	7.18	12.42	11.89	12.17	11.88	10.24	3.54	22.92	11.51	4.10	2.52	4.78	121.01
Upper Zakum	4.30	3.74	7.37	8.31	8.27	8.34	8.20	3.30	46.34	12.91	7.60	6.72	19.11	144.51
WTI Light	5.96	10.23	16.54	12.24	10.21	9.66	8.29	2.89	21.81	9.57	4.43	3.16	4.64	119.63

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Note: These entries represent the cutting-edge proceedings and peer-reviewed studies as of the current 2026 academic cycle.

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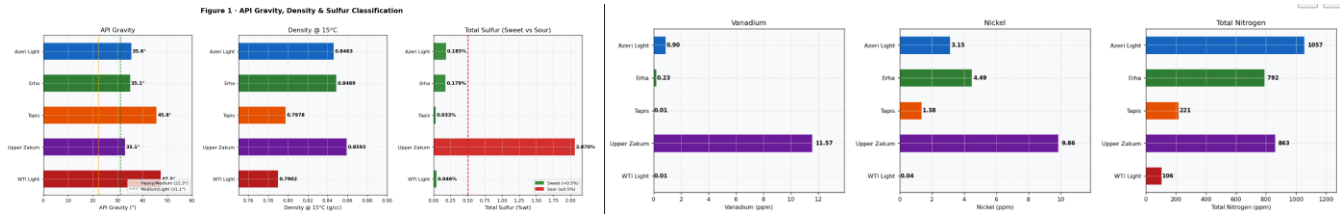

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